

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10		53, 381								810	NIPPLE TO BE CUT FROM SCH 160 PIPE.			
20										B1	ALL BOLTS ARE METRIC DIA.			
20										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
30							*****VALVES*****							
30		VA5					***** MODULAR VALVES *****			B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22			
40														
40			MODULAR DBB	K01AM10	0.75	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0					
50		VAV	MODULAR DBB	K03AM10	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.			
60			MODULAR DBB	M01AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES			
70			MODULAR DBB	M03AM10	0.75	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.			
80		VAW	MODULAR DBB	M03EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.			
90		VAW	MODULAR DBB	M03EM10	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.			
100		VB2					***** BALL VALVES *****							
100														
110														
110			BALL	A01AM09	3	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS			
120			BALL	A01AR09	4	6	BALL 150# RF LTCS + CLAD INC 825 INC 625 TRIM FLOATING SOFT RB LO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
130			BALL	A01AR15	8	8	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
140			BALL	B01AM09	0.75	2	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
150		VA5	BALL	B01AM09	3	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
160		VA5	BALL	B01AR11	4	6	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0	VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.			
170		VA5	BALL	B01AR15	8	8	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT FB GO	0	0	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.			
180		FS1	BALL	B03AM09	0.75	2	BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0	53	USE OF NPS 1/2 IS RESTRICTED TO ORIFICE FLANGE TAP CONNECTIONS.			
190		FS1	BALL	A03AM09	3	3	BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0	381	ALL ALLOY 825 MATERIALS SHOULD FOLLOW REQUIREMENTS IN CPM-SU-5013. ALL ALLOY 825 MATERIALS SHALL BE STABILIZED ANNEALED.			
200							***** BUTTERFLY VALVES *****							
200														
210			BUTTERFLY	T01AR63	10	12	BUTTERFLY 150# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO	0	0					
220														
220		VQ2					***** GATE VALVES *****							
230														
230			GATE	G01AM49	0.75	3	GATE 150# RF INC 825 INC 625 TRIM WEDGE HW	0	0					
240			GATE	G01AR43	4	8	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE HW	0	0					
250		VA5	GATE	G01AR43	10	10	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE HW	0	0					
260		VA5	GATE	G01AR44	12	12	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE GO	0	0					
270							***** GLOBE VALVES *****							
270														
280			Globe	L01AM49	0.75	3	GLOBE 150# RF INC 825 INC TRIM 625 OS&Y HW	0	0					
290			Globe	L01AR49	2	3	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW	0	0					
300			Globe	L01AR44	8	10	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y GO	0	0					
310							***** CHECK VALVES *****							
310														
320		VC1	Check	C01AM51	0.75	1.5	CHECK PISTON 150# RF INC 825 INC TRIM 625	0	0					
330		VC2	Check	C01AM46	2	3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED	0	0					
340		VC2	Check	C01AR46	4	6	CHECK DUAL WAFER LUG 150# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED	0	0					
350		VC2	Check	C01AR53	8	24	CHECK DUAL DBL FLG 150# RF LTCS + CLAD INC 825 INC TRIM 625	0	0					
360							***** PIPE *****							
360														
370														
370			PIPE	5732001-CD	0.75	1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED PE	0	0					
380			PIPE	5732002-CD	1.5	1.5	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED PE	0	0					
390			PIPE	5361603-CD	2	6	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED BE	0	0					
400			PIPE	5361603-WLD20	8	12	PIPE SCH 20 WLD ALLOY 825 B474 UNS N08825 CL 3 STABILIZED ANNEALED BE	0	0					
410		810	PIPE	5361608-CD	0.5	0.75	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED PE	0	0					
420							***** FITTINGS *****							
420														
430														
430			ELBOW	5425111	0.75	1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPNICMC-S	0	0					
440			ELBOW	5424722	1.5	6	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
450			ELBOW	5424722-WLD20	8	12	ELL 90 DEG LR SCH 20 ALLOY 825 B366-WPNICMC-WX	0	0					
460			ELBOW	5425113	0.75	1	ELL 45 DEG LR SCH 80S ALLOY 825 B366-WPNICMC-S	0	0					
470			ELBOW	5424727	1.5	6	ELL 45 DEG LR SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
480			ELBOW	5424727-WLD20	8	12	ELL 45 DEG LR SCH 20 ALLOY 825 B366-WPNICMC-WX	0	0					
490														
490			Tee	5425119	0.75	1	TEE SCH 80S ALLOY 825 B366-WPNICMC-S	0	0					
500			Tee	5424725	1.5	6	TEE SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
			TENGIZCHEVROIL TCO, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	
					ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 3.8 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO		MATERIAL: ALLOY 825 FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 230 °C		12/16/2024	
													BY OZAC	
													REV U04	
													SHEET Page 1 of 3	
													PIPING LINE CLASS 150A80	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
510			Tee	5424725-WLD20	8	12	TEE SCH 20 ALLOY 825 B366-WPNICMC-WX	0	0					
520			TEE RED	5425126	0.75	0.75	TEE RED SCH 80S ALLOY 825 B366-WPNICMC-S	1	1					
530			TEE RED	5425124	0.75	1	TEE RED SCH 40S/80S ALLOY 825 B366-WPNICMC-S	1.5	2					
540			TEE RED	5424724	1.5	4	TEE RED SCH 40S ALLOY 825 B366-WPNICMC-S	2	6					
550			TEE RED	5424724-W2040	4	6	TEE RED SCH 20/40S ALLOY 825 B366-WPNICMC-WX	8	12					
560			TEE RED	5424724-WLD20	8	10	TEE RED SCH 20 ALLOY 825 B366-WPNICMC-WX	10	12					
570														
570			Cap	5424270	0.75	1	CAP SCH 80S ALLOY 825 B366-WPNICMC-S	0	0					
580			Cap	5424219	1.5	6	CAP SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
590			Cap	5424219-20	8	12	CAP SCH 20 ALLOY 825 B366-WPNICMC-S	0	0					
600														
600			RED CONC	5425122	0.75	0.75	REDUCER CONC SCH 80S ALLOY 825 B366-WPNICMC-S	1	1					
610			BOLTING	5425143	0.75	1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPNICMC-S	1.5	2					
620			RED CONC	5424733	1.5	4	REDUCER CONC SCH 40S ALLOY 825 B366-WPNICMC-S	2	6					
630			RED CONC	5424733-W2040	4	6	REDUCER CONC SCH 20/40S ALLOY 825 B366-WPNICMC-WX	8	12					
640			RED CONC	5424733-WLD20	8	10	REDUCER CONC SCH 20 ALLOY 825 B366-WPNICMC-WX	10	12					
650			RED ECC	5425123	0.75	0.75	REDUCER ECC SCH 80S ALLOY 825 B366-WPNICMC-S	1	1					
660			RED ECC	5425145	0.75	1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPNICMC-S	1.5	2					
670			RED ECC	5424734	1.5	4	REDUCER ECC SCH 40S ALLOY 825 B366-WPNICMC-S	2	6					
680			RED ECC	5424734-W2040	4	6	REDUCER ECC SCH 20/40S ALLOY 825 B366-WPNICMC-WX	8	12					
690			RED ECC	5424734-WLD20	8	10	REDUCER ECC SCH 20 ALLOY 825 B366-WPNICMC-WX	10	12					
700														
700							***** OLETS *****							
710														
710			OLETS	5504121	0.75	1	NIPOFLANGE 150# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2	6					
720			OLETS	5504122	1.5	2	NIPOFLANGE 150# RF SCH 40S ALLOY 825 B564 UNS N08825	2	6					
730			OLETS	5504150	0.75	1	NIPOFLANGE 150# RF SCH 20/80S ALLOY 825 B564 UNS N08825	8	12					
740			OLETS	5504155	1.5	2	NIPOFLANGE 150# RF SCH 20/40S ALLOY 825 B564 UNS N08825	8	12					
750		FS1	OLETS	5513491	0.75	1	NIPOFLANGE 300# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2	6					
760		FS1	OLETS	5513495	1.5	2	NIPOFLANGE 300# RF SCH 40S ALLOY 825 B564 UNS N08825	2	6					
770		FS1	OLETS	5513491-2080	0.75	1	NIPOFLANGE 300# RF SCH 20/80S ALLOY 825 B564 UNS N08825	8	12					
780		FS1	OLETS	5513495-2040	1.5	2	NIPOFLANGE 300# RF SCH 20/40S ALLOY 825 B564 UNS N08825	8	12					
790		UJ2	OLETS	5513479-40160	1.5	1.5	NIPOFLANGE 300# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	4	6					
800		UJ2	OLETS	5513420	1.5	1.5	NIPOFLANGE 300# RF SCH 20/160 ALLOY 825 B564 UNS N08825	8	12					
810		FF5	OLETS	5504198	0.75	1	NIPOFLANGE FLAT BOTTOM 150# RF SCH 80S ALLOY 825 B564 UNS N08825	0	0					
820		FF5	OLETS	5504199	1.5	1.5	NIPOFLANGE FLAT BOTTOM 150# RF SCH 40S ALLOY 825 B564 UNS N08825	0	0					
830		FF5	OLETS	5504518	0.75	1	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S ALLOY 825 B564 UNS N08825	0	0					
840		FF5	OLETS	5504519	1.5	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 40S ALLOY 825 B564 UNS N08825	0	0					
850														
850			OLETS	5513437	0.75	1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2	6					
860			OLETS	5513442	0.75	1	WELDOLET SCH 20/80S ALLOY 825 B564 UNS N08825	8	12					
870			OLETS	5513411	1.5	2	WELDOLET SCH 40S ALLOY 825 B564 UNS N08825	4	6					
880			OLETS	5513441	1.5	4	WELDOLET SCH 20/40S ALLOY 825 B564 UNS N08825	8	12					
890														
890		UJ2	ELBOFLG	5516626-40160	1.5	1.5	ELBOFLANGE 300# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	4	6					
900		UJ2	ELBOFLG	5516629	1.5	1.5	ELBOFLANGE 300# RF SCH 20/160 ALLOY 825 B564 UNS N08825	8	12					
910														
910		FS1					***** FLANGES *****							
920														
920		FA8	FLG RED	5614311	0.5	0.5	FLG SCRD (FNPT) REDUCING 150# RF ALLOY 825 B564 UNS N08825	0.75	0.75					
930		FA8	FLG RED	5614310	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF ALLOY 825 B564 UNS N08825	0.75	0.75					
940			Flange	5569030	0.5	1	FLG WN 150# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0					
950			Flange	5569026	1.5	6	FLG WN 150# RF ALLOY 825 SCH 40S B564 UNS N08825	0	0					
960			Flange	5569276-20	8	12	FLG WN 150# RF ALLOY 825 SCH 20 B564 UNS N08825	0	0					
970		FS1	Flange	5569290	0.5	1	FLG WN 300# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0					
980		FS1	Flange	5569276	1.5	6	FLG WN 300# RF ALLOY 825 SCH 40S B564 UNS N08825	0	0					
990		FS1	Flange	5569276-20	8	12	FLG WN 300# RF ALLOY 825 SCH 20 B564 UNS N08825	0	0					
1000			Blind	5617150	0.5	12	FLG BLIND 150# RF ALLOY 825 B564 UNS N08825	0	0					
1010		FS1	Blind	5617159	0.5	12	FLG BLIND 300# RF ALLOY 825 B564 UNS N08825	0	0					
1020														
1020							***** ORIFICE FLANGES *****							
1030														
1030			ORIFICE FLG	5650830	2	6	WN ORIF SET 300# RF ALLOY 825 B564 UNS N08825 SCH 40S	0	0					
1040			ORIFICE FLG	5650830-20	8	12	WN ORIF SET 300# RF ALLOY 825 B564 UNS N08825 SCH 20	0	0					
1050														
1050							***** NIPPLES *****							
1060														
1060							NIPPLE CUT FROM PIPE							
1070														
1070							***** GASKETS *****							
1080														
1080			GASKET	5673653	0.5	12	GASKET 150# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0					
1090		FS1	GASKET	5673657	0.5	12	GASKET 300# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0					
			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
			ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 3.8 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: ALLOY 825 FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 230 °C		12/16/2024	BY OZAC	U04	Page 2 of 3	150A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1100							***** BOLTING *****			
1100										
1110										
1110		B1	BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1120										
1120		SB6	BOLTING	5675675	12	12	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1130										
1130							***** BLINDS & SPACERS *****			
1140										
1140			SPEC BLIND	5749681	0.75	10	SPECTACLE BLIND 150# RF ALLOY 825 B424 UNS N08825	0	0	
1150										
1150			PADDLE	5750231-150	12	12	PADDLE BLANK 150# RF ALLOY 825 B424 UNS N08825	0	0	
1160			PADDLE	5750232-150	12	12	PADDLE SPACER 150# RF ALLOY 825 B424 UNS N08825	0	0	
1170										
1170		FS1	SPEC BLIND	5749683	0.75	10	SPECTACLE BLIND 300# RF ALLOY 825 B424 UNS N08825	0	0	
1180										
1180			PADDLE	5750231-300	12	12	PADDLE BLANK 300# RF ALLOY 825 B424 UNS N08825	0	0	
1190			PADDLE	5750232-300	12	12	PADDLE SPACER 300# RF ALLOY 825 B424 UNS N08825	0	0	
1200										
1210										
1210							USE BRANCH CHART 300A80			
1220										
1220							PIPE BENDS ----- NONE			
1230										
1230							THREAD SEALANT-----PTFE PASTE			

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 3.8 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: ALLOY 825 FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 3 of 3	150A80